Owen Latt

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Education

University of Pittsburgh — Pittsburgh, PA
Bachelor of Science in Computer Science, Minor in Business — Aug. 2024 – April 2028

3.88 GPA | Dean's List

Relevant Coursework: Data Structures & Algorithms 1/2, Computer Organization & Assembly Language, Intro to Systems and Software, Intermediate Programming, Statistics (R, data manipulation & visualization), Discrete Math, Linear Algebra, Social Implications of Computing Technology,

In Progress (Fall 2026): Software Engineering, Math Foundations for Machine Learning (Pandas, NumPy, Matplotlib, Seaborn)

Quakertown Community High School — Quakertown, PA
4.02 GPA, 45/414 Class rank — Aug. 2020 – May 2024

Relevant Coursework: AP Computer Science Principles, AP Computer Science A

Technical Skills

Languages: Java, Python, C, JavaScript, HTML/CSS, R

Libraries/Frameworks: Pandas, NumPy, Matplotlib, Seaborn, GeoPandas, Shapely, Tkinter

Concepts: Data Structures & Algorithms, Operating Systems (processes, system calls, signal handling), Computer Architecture, Data Manipulation & Cleaning, Data Visualization, Object-Oriented Programming

Tools/Software: Git/GitHub, VS Code, Terminal/SSH, GDB, R Studio, Video Editing

Projects

Best Neighborhood in Pittsburgh | *Python (Pandas, GeoPandas, Shapely)* — Team Project

- Analyzed three public datasets (crime, parks, schools) to rank Pittsburgh's roughly 90 neighborhoods on safety, recreation, and education, using GeoPandas and Shapely to spatially join school coordinates to neighborhood boundaries
- Cleaned and merged datasets with inconsistent formatting and incomplete coverage, then built a composite ranking system to identify the top neighborhood for families

Custom Unix Shell | *C, Linux System Calls* — Solo Project

- Built a Unix shell in C that parses user input and executes built-in commands (`cd`, `exit`) and external programs using `fork()`, `execvp()`, and `waitpid()`, reporting exit status and signal-based termination
- Implemented I/O redirection (<, >) via `freopen()` with input validation, and configured signal handling so the shell ignores `SIGINT` while child processes retain default behavior, preventing Ctrl+C from killing the shell itself

Evade | *Python (Tkinter), JavaScript/HTML5 Canvas* — Solo Project

- Built a real-time survival game in Python (Tkinter) and ported it to the browser using HTML5 Canvas, implementing circle-based collision detection and a scaling difficulty system linking enemy size inversely to speed
- Refactored the web version's game loop to use delta-time scaling with `requestAnimationFrame`, decoupling gameplay speed from monitor refresh rate so difficulty stays consistent across devices, and added persistent high score tracking via `localStorage`

Work Experience

Lees Hoagie House (Shift Manager) — Dec. 2021– Present
Quakertown, PA

- Independently manage store operations, overseeing workflow and customer service during assigned shifts
- Trusted to track inventory and order stock, handle and reconcile cash, and open/close the store securely

Additional Activities

- **YouTube Channel** – Created and edited gaming video editing tutorials; grew to 3,000+ subscribers and 500,000+ views
- **Robotics Club** (High School) – Collaborated with a student team to design and build competition robots